SIMATS ENGINEERING
ASSESSMENT TOOL 3: Comparative Analysis

COURSE NAME: COMPUTER VISION **COURSE CODE: ITA05**

COURSE OUTCOME COVERED

CO4: *Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics.* (BTL 5)

Assessment Tool Weightage: 10%

Compute the required performance measures from the given data and compare the techniques based on the results. Evaluate and justify your conclusion using the calculated values.

4. Depth Estimation Comparison (Stereo vs SfM)
Estimated depth values (in meters):

Object	Stereo Vision	Structure from Motion
A	2.0	2.5
B	3.0	3.8
C	4.0	5.0

Actual depths: A=2, B=3, C=4
(a) Compute absolute error for both methods.
(b) Compare and evaluate which method is more accurate.

Depth Estimation Comparison — Stereo Vision vs SfM

Given Data

Object	Actual Depth (m)	Stereo Vision (m)	Structure from Motion (SfM) (m)
A	2.0	2.0	2.5
B	3.0	3.0	3.8
C	4.0	4.0	5.0

(a) Absolute Error

The formula for absolute error is:

Absolute Error=|Estimated Depth−Actual Depth|

Stereo Vision

- **Object A:** (|2.0-2.0| = 0.0) m
- **Object B:** (|3.0-3.0| = 0.0) m
- **Object C:** (|4.0-4.0| = 0.0) m

Mean Absolute Error (MAE):

$$MAE = 0+0+0/3 = 0m$$

Structure from Motion (SfM)

- **Object A:** (|2.5-2.0| = 0.5) m
- **Object B:** (|3.8-3.0| = 0.8) m
- **Object C:** (|5.0-4.0| = 1.0) m

Mean Absolute Error (MAE):

Results

M A E = 0.5+0.8+1.0/3

M A E = 0.77 m

Object	Actual (m)	Stereo Error (m)	SfM Error (m)
A	2.0	**0.0**	0.5
B	3.0	**0.0**	0.8
C	4.0	**0.0**	1.0
MAE	—	**0.00 m**	**0.77 m**

(b) Comparison and Evaluation

Stereo Vision has an absolute error of **0 m for all three objects**, resulting in a **MAE of 0 m**. This means its estimated depths exactly match the actual depths.

SfM produces errors of **0.5 m, 0.8 m, and 1.0 m**, with a **MAE of approximately 0.77 m**. Therefore, it is less accurate for the given dataset.

Conclusion

→ Stereo Vision is the more accurate technique.

The calculated values clearly support this decision:

- **Stereo Vision MAE = 0 m**
- **SfM MAE = 0.77 m**

Since a lower absolute error indicates higher depth-estimation accuracy, Stereo Vision performs significantly better than SfM for the given objects. Therefore, Stereo Vision should be selected when accurate depth measurement is the primary requirement.

8. **Optical Flow Consistency**

Motion values:

Frame	Method A	Method B
1	4	6
2	5	8
3	6	9

(a) Compute average motion.

(b) Compare consistency and evaluate which is more stable.

OPTICAL FLOW CONSISTENCY

Given Data

Frame	Method A	Method B
1	4	6
2	5	8
3	6	9

(a) Compute Average Motion

The average motion is:

$$\text{Average} = \frac{\text{Sum of motion values}}{\text{Number of frames}}$$

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Method A

$$\text{Average}_A = \frac{4+5+6}{3} = \frac{15}{3} = \mathbf{5}$$

Method B

$$\text{Average}_B = \frac{6+8+9}{3} = \frac{23}{3} = \mathbf{7.67}$$

(b) Compare Consistency

To evaluate consistency, observe how much the motion values vary across frames.

- **Method A:** 4 → 5 → 6
  Variation = $6 - 4 = \mathbf{2}$
- **Method B:** 6 → 8 → 9
  Variation = $9 - 6 = \mathbf{3}$

Method A has a smaller variation, meaning its motion estimates change less between frames.

Measure	Method A	Method B
Average motion	**5.00**	7.67
Range	**2**	3
Stability	**Higher**	Lower

Conclusion

→ **Method A is more stable and consistent.**

Although **Method B has a higher average motion (7.67)**, its values fluctuate more across frames. Method A has a lower variation and therefore provides **more consistent optical-flow measurements**.

Final evaluation: *For applications where motion stability and consistency are more important than the magnitude of detected motion, Method A is the better choice.*

9. **Detection Accuracy Comparison**

Results:

Model	**Correct**	**Total**
A	180	200
B	170	200

(a) Compute accuracy.

(b) Compare and evaluate.

DETECTION ACCURACY COMPARISON

Given Data

Model	Correct Detections	Total Samples
A	180	200
B	170	200

(a) Compute Accuracy

The formula is:

$$\text{Accuracy} = \frac{\text{Correct Predictions}}{\text{Total Predictions}} \times 100$$

Model A

$$\text{Accuracy}_A = \frac{180}{200} \times 100 = \mathbf{90\%}$$

Model B

$$\text{Accuracy}_B = \frac{170}{200} \times 100 = \mathbf{85\%}$$

(b) Comparison and Evaluation

Model	Accuracy
A	**90%**
B	85%

Model A achieves **90% accuracy**, while Model B achieves **85% accuracy**.

The difference is:

90%−85%=5 percentage points

Therefore, **Model A is more accurate** and performs better for the given detection task.

Conclusion

→ **Model A should be selected.**

Justification: *Model A correctly detects 180 out of 200 samples, achieving 90% accuracy. Model B achieves only 85%. Since Model A has a 5-percentage-point accuracy advantage, it provides more reliable detection and is the better-performing technique.*

10. **Error Rate Comparison**

Results:

Method	**Errors**	**Total**
A	20	100
B	10	100

(a) Compute error rate.

(b) Compare and evaluate.

ERROR RATE COMPARISON

Given Data

Method	Errors	Total
A	20	100
B	10	100

(a) Compute Error Rate

The formula is:

$$\text{Error Rate} = \frac{\text{Number of Errors}}{\text{Total Cases}} \times 100$$

Method A

$$\text{Error Rate}_A = \frac{20}{100} \times 100 = \mathbf{20\%}$$

Method B

$$\text{Error Rate}_B = \frac{10}{100} \times 100 = \mathbf{10\%}$$

(b) Comparison and Evaluation

Method	Error Rate
A	20%
B	10%

Method B has an error rate of **10%**, which is **10 percentage points lower** than Method A's 20%. Since a **lower error rate indicates better performance**, Method B is more reliable and accurate.

Conclusion

→ **Method B should be selected.**

Justification: *Method B produces only 10 errors out of 100 cases, compared with 20 errors for Method A. Its error rate is therefore 50% lower relative to Method A, making Method B the better-performing technique.*

11. **Motion Estimation Variance**

Errors:

Frame	A	B
1	2	6
2	3	7
3	2	5

(a) Compute mean error.

(b) Compare variability and evaluate stability.

MOTION ESTIMATION VARIANCE

Given Data

Frame	Method A	Method B
1	2	6
2	3	7
3	2	5

(a) Compute Mean Error

The mean error is:

$$\text{Mean Error} = \frac{\text{Sum of Errors}}{\text{Number of Frames}}$$

Method A:

$$\frac{2+3+2}{3} = \frac{7}{3} = \mathbf{2.33}$$

Method B:

$$\frac{6+7+5}{3} = \frac{18}{3} = \mathbf{6}$$

(b) Compare Variability and Evaluate Stability

To compare stability, we calculate the **variance**. Lower variance means the errors fluctuate less and the method is more stable.

Using population variance:

Method A

Mean = 2.33

$$Variance_A = \frac{(2-2.33)^2 + (3-2.33)^2 + (2-2.33)^2}{3}$$

$$\mathbf{Variance_A \approx 0.22}$$

Method B

Mean = 6

$$Variance_B = \frac{(6-6)^2 + (7-6)^2 + (5-6)^2}{3}$$

$$\mathbf{Variance_B \approx 0.67}$$

Comparison

Measure	**Method A**	**Method B**
Mean Error	**2.33**	6.00
Variance	**0.22**	0.67
Stability	**Higher**	Lower

Conclusion

→ Method A is more stable and performs better.

Method A has both a **lower mean error (2.33)** and **lower variance (0.22)** compared with Method B's mean error of **6.00** and variance of **0.67**.

Justification: *A lower mean error indicates better estimation accuracy, while a lower variance indicates greater consistency. Therefore, Method A is both more accurate and more stable than Method B.*

12. **IOU Comparison**

IOU values:

Object	**A**	**B**
1	0.6	0.8
2	0.7	0.85
3	0.65	0.9

(a) Compute average IOU.

(b) Compare and evaluate detection quality.

IOU COMPARISON

Given Data

Object	Method A	Method B
1	0.60	0.80
2	0.70	0.85
3	0.65	0.90

(a) Compute Average IoU

The average IoU is:

$$\text{Average IoU} = \frac{\text{Sum of IoU values}}{\text{Number of Objects}}$$

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Method A

$$\text{Average IoU}_A = \frac{0.60 + 0.70 + 0.65}{3}$$

$$= \frac{1.95}{3} = \mathbf{0.65}$$

Method B

$$\text{Average IoU}_B = \frac{0.80 + 0.85 + 0.90}{3}$$

$$= \frac{2.55}{3} = \mathbf{0.85}$$

(b) Compare and Evaluate Detection Quality

Method	Average IoU	Detection Quality
A	0.65	Moderate
B	**0.85**	**High**

IoU (Intersection over Union) measures how closely the predicted bounding box overlaps with the actual object region. **Higher IoU indicates better localization and detection quality.**

Method B achieves an average IoU of **0.85**, compared with **0.65** for Method A.

Difference:

$0.85 - 0.65 = 0.20$

Conclusion

→ **Method B provides better detection quality.**

Justification: *Method B achieves an average IoU of 0.85, which is 0.20 higher than Method A's 0.65. Therefore, Method B produces better bounding-box overlap and more accurate object localization.*

15. **Recognition Precision Comparison**

Results:

System	**TP**	**FP**
A	90	10
B	85	5

(a) Compute precision.

(b) Compare and evaluate performance.

RECOGNITION PRECISION COMPARISON

Given Data

System	True Positives (TP)	False Positives (FP)
A	90	10
B	85	5

(a) Compute Precision

The formula for precision is:

$$\text{Precision} = \frac{TP}{TP + FP} \times 100$$

System A

$$\text{Precision}_A = \frac{90}{90 + 10} \times 100$$

$$= \frac{90}{100} \times 100 = \mathbf{90\%}$$

System B

$$\text{Precision}_B = \frac{85}{85 + 5} \times 100$$

$$= \frac{85}{90} \times 100 = \mathbf{94.44\%}$$

(b) Compare and Evaluate Performance

System	**TP**	**FP**	**Precision**
A	90	10	90%
B	85	5	**94.44%**

System B has fewer true positives than System A, but it also produces significantly fewer false positives.

The precision improvement is:

94.44%−90%=4.44 percentage points

Conclusion

→ System B performs better in terms of recognition precision.

Justification: *System B achieves 94.44% precision compared with 90% for System A. Although System A detects more true positives, System B generates fewer false positives. Therefore, when the goal is to minimize incorrect positive recognitions, System B is the better-performing system.*